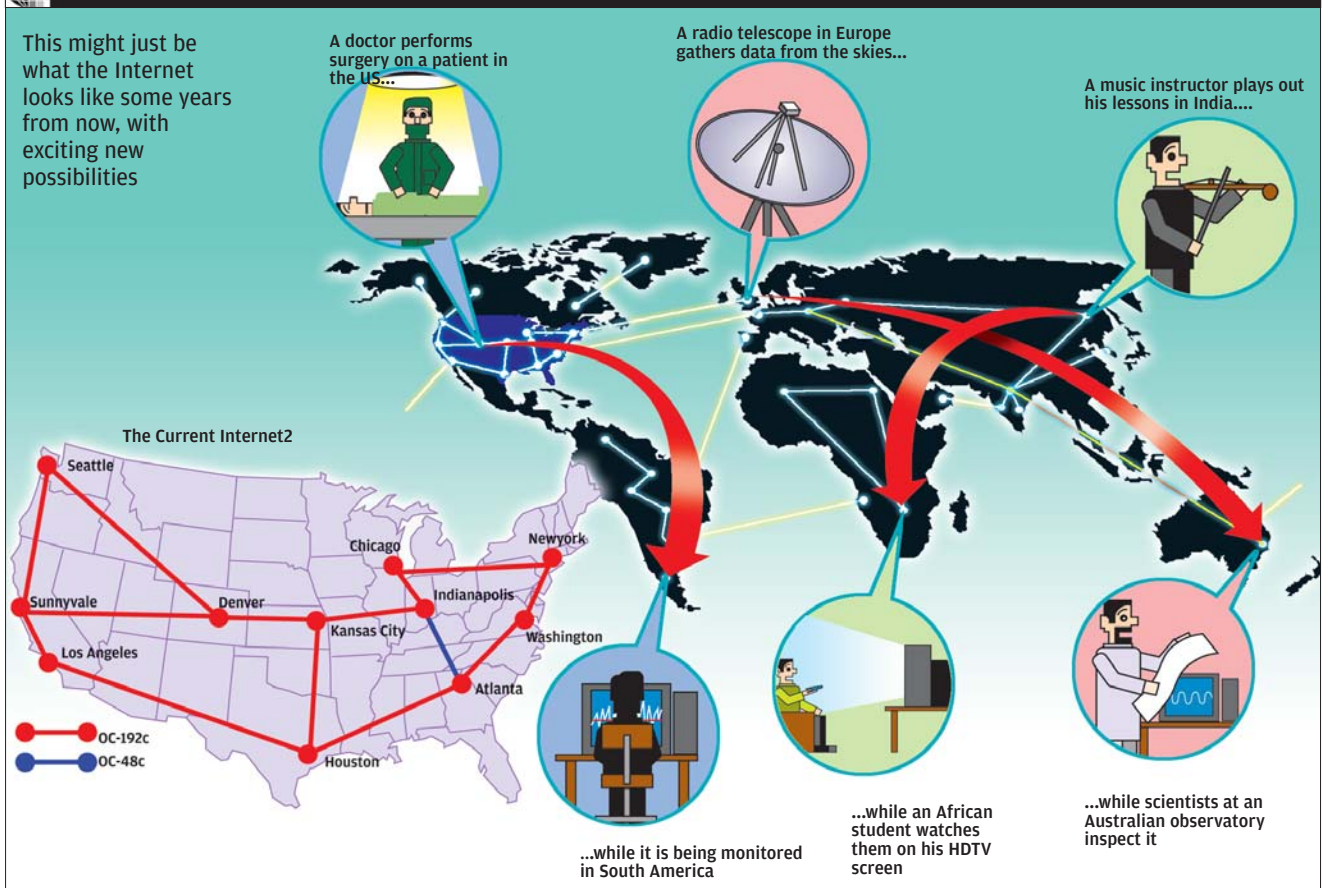




Is Internet2 A Harbinger Of These Possibilities?



Infograph: Solomon Lewis

Internet Protocol—IPv4—supports datagrams of up to 64 KB.)

Important considerations in IPv6 are the “flow label” and the “priority setting.” Both these features improve the Quality of Service. Using the flow label and priority, you could indicate to a router that a video stream needs to go at no more than a particular jitter level. The router could also be instructed to optimise transmission and improve efficiency. These two new fields in the IPv6 header will become more and more important as we move toward ultrahigh-bandwidth applications, such as those available on I2.

Possibilities

I2 is used for applications not possible with the limited speeds and weak infrastructure of today’s Internet. Such uses include education, research, and the like. A music teacher in New York could make a live lesson available via I2 to students in Los Angeles, thousands of miles away. The audio and video would be DVD-quality, in real-time. Astronomers at the University of Florida could observe data in real-time from a telescope at Palo Alto, California.

In fact, it’s already happening. In the near future, we can envisage telemedicine happening via the I2-like Internet. Indian doctors could monitor a surgery in Brazil, and perhaps even perform actions that have a real-time effect in the operating theatre.

This is just an indication of what is possible with I2 and its probable successor, which will

With Internet2, a music teacher in New York could give live lessons to students in Los Angeles. It can be that fast and useful

be nothing short of a total revolution in connectivity—with no speed barriers. I2 is just showing us the way.

IPv6 comes in here as well. When such high bandwidths are involved, one would like to reduce the wastage that happens with “unicast”—data being sent from one node to another. Multicast is where a specific list of addresses receives the data, without it having to be sent to each node once. IPv4 does have multicast capabilities, but IPv6 improves on this, in terms of flexibility of addressing.



Abilene, Mean, Stronger

The Abilene network is a high-performance backbone network that enables the development of advanced Internet applications, and the deployment of leading-edge network services by I2 universities and research laboratories across the US. Abilene was created by the I2 community, and connects regional network aggregation points—known as gigaPoPs—to provide advanced network capabilities to over 220 I2 university, corporate, and affiliate member institutions. The current network is primarily a 10 Gbps backbone, employing optical transport technology and advanced high-performance routers. Abilene is a partnership effort between Internet2, Qwest Communications, Nortel Networks, Juniper Networks, and Indiana University.